DATA STRUCTURES AND ALGORITHMS – 3 | 25CS2103E

Resume–Job Matching & Talent-Marketplace Engine

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Abstract

The Resume–Job Matching & Talent-Marketplace Engine automates and optimizes recruitment by efficiently matching large collections of resumes with relevant job postings. It tackles challenges such as inconsistent skill terminology, varying candidate qualifications, and the need for optimal workforce allocation.

The system indexes resumes and job descriptions using efficient data structures, normalizes skill vocabularies, and computes candidate–job compatibility scores based on skills, experience, qualifications, and role requirements. Optimization algorithms then assign candidates to suitable roles, maximizing overall match quality while minimizing mismatches. The engine also determines the minimum set of skills needed to staff a given team, helping identify skill gaps and improve workforce planning.

The project demonstrates practical application of hashing, graphs, string matching, sorting, searching, greedy algorithms, and bipartite matching/optimization — offering a scalable, data-driven approach to recruitment and team formation.

Introduction

The Problem

Recruiters must sift through large volumes of resumes and job postings, where skills are described inconsistently and candidate qualifications vary widely, making manual matching slow and error-prone.

Our Approach

The system represents resumes and jobs using efficient data structures, normalizes skill terms, and scores candidate job fit using multiple weighted factors.

Core Technique

Compatibility is modeled as a weighted bipartite graph between candidates and roles, solved with optimization methods such as the Hungarian algorithm to maximize overall assignment quality.

Added Value

Beyond matching, the engine identifies the minimum skill set required to staff a team, surfacing skill gaps for better workforce planning.

Methodology

1

Data collection

Collect and parse resumes and job postings.

2

Skill Extraction

Identify skills, experience, and qualifications.

3

Compatibility Scoring

Score candidate–job fit on skills, experience & qualifications

4

Candidate Ranking

Rank candidates using *Sorting & Priority Queue*.

5

Optimal Assignment

Assign candidates to roles using *Bipartite Matching*.

6

Gap Analysis

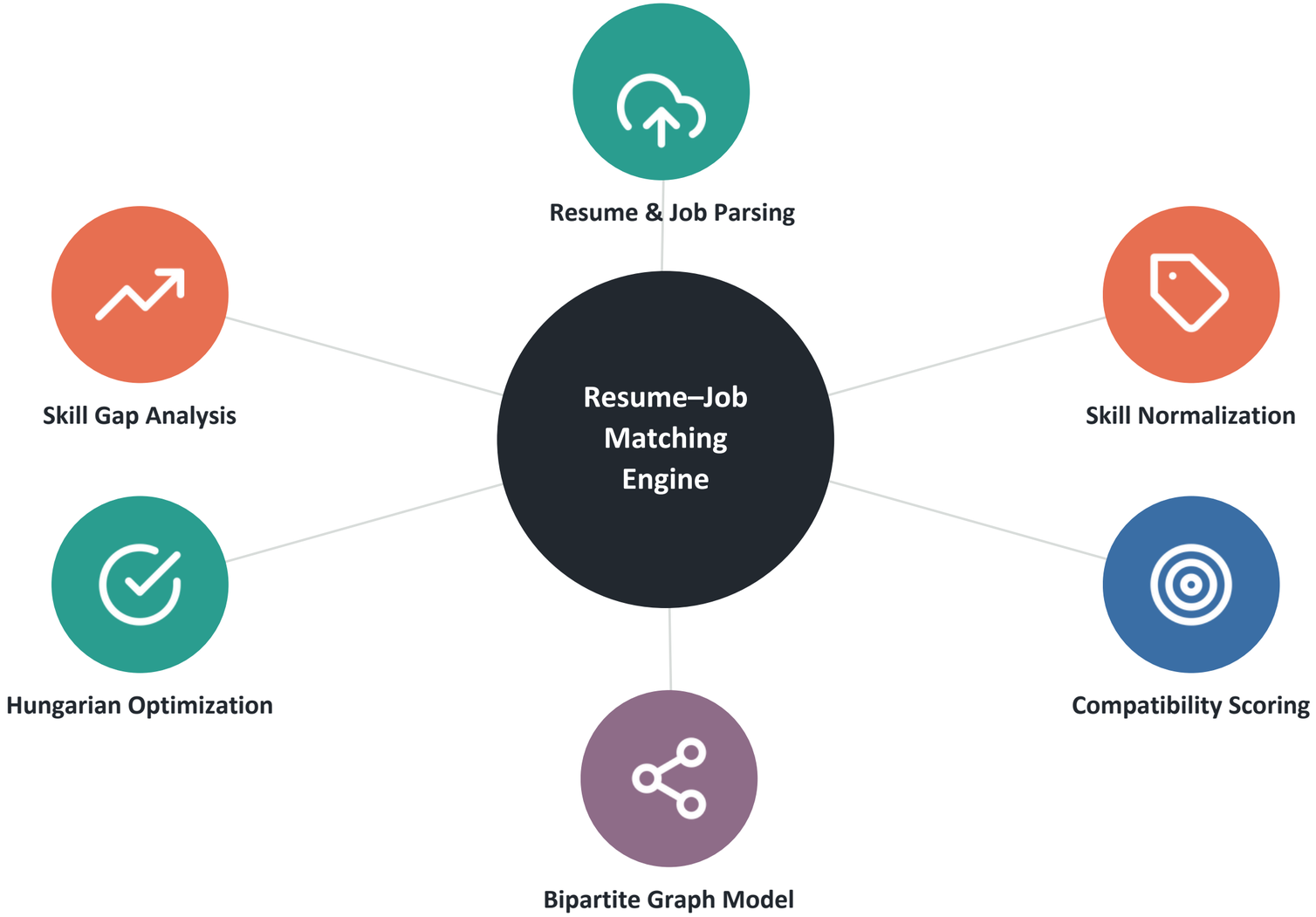
Determine the minimum skills needed to staff a team

Why this order matters

Normalizing skills before scoring avoids false mismatches from wording alone; framing the problem as a weighted bipartite graph lets a proven optimization method (Hungarian algorithm) guarantee the best overall assignment rather than a greedy, locally-good one.

Pipeline: raw input → normalized skills → scored compatibility → optimized assignment → workforce insight

Project Mind Map



Literature Survey

Resume Parsing & NLP

Extracting structured skills and experience from free-form resume text.

Bipartite Graph Matching

Modeling candidates and roles as a weighted bipartite graph for matching.

Assignment Optimization

Hungarian algorithm and related methods for optimal one-to-one allocation.

Job Recommendation Systems

Ranking and suggesting roles based on candidate profile similarity.

Software Requirements

Programming Language	Python / Java / C++
Development Environment	VS Code / PyCharm / Eclipse
Libraries & Tools	Data structure & graph libraries, NetworkX (or equivalent)
Version Control	Git & GitHub
Operating System	Windows / Linux / macOS
Documentation	MS Word / LaTeX for reporting

Hardware Requirements

Processor	Intel i5 / AMD Ryzen 5 or higher
RAM	8 GB minimum (16 GB recommended)
Storage	256 GB SSD or higher
Display	1080p monitor for development & demo
Internet	Stable connection for version control & testing